3.3 Shutting down a virtual machine

To avoid incurring charges, you should always turn off virtual machines when you're not using them. You can use the following four actions to control a virtual machine's state:

- *Start*—You can always start a stopped virtual machine. If you want to create a completely new machine, you'll need to launch another virtual machine.
- *Stop*—You can always stop a running virtual machine. A stopped virtual machine doesn't incur charges, except for attached resources like network-attached storage. A stopped virtual machine can be started again but likely on a different host. If you're using network-attached storage, your data persists.
- *Reboot*—Have you tried turning off your virtual machine, then turning it on again? If you need to reboot your virtual machine, this action is what you want. You won't lose any persistent data when rebooting a virtual machine because it stays on the same host.
- *Terminate*—Terminating a virtual machine means deleting it. You can't start a virtual machine that you've terminated. The virtual machine is deleted, usually together with its dependencies, like network-attached storage and public and private IP addresses. A terminated virtual machine doesn't incur charges.

WARNING The difference between *stopping* and *terminating* a virtual machine is important. You can start a stopped virtual machine. This isn't possible with a terminated virtual machine. If you terminate a virtual machine, you delete it.

Figure 3.19 illustrates the difference between stopping and terminating an EC2 instance, with the help of a flowchart.